

Day 14 lab: capstone proposal and start of the build

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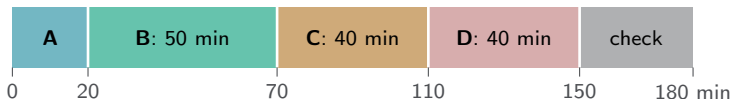
- ▶ **Goal:** an approved capstone proposal and a started build.
- ▶ **Boards:** the XIAOML Kit and the Raspberry Pi 5 of both lab groups of your team.
- ▶ **Deliverable:** the approved proposal `proposal.md`, and the file `report.md` with the budgets, the result of the review, the result of the first test, and the Decision Log.

From the lecture to the lab

Part 3 of the lecture gave a design method in seven steps, the budgets for latency, power, and cost, and the questions of a design review. Today your team applies them to its own system. `Docs/capstone.md` is the capstone brief.

Key point: Test the riskiest assumption today. A plan that fails on Day 14 can still change.

Plan of the three hours



Part	Work	Time
A	Teams: form the team, select a problem	20 min
B	Proposal: requirements, boards, model, data, metrics, budgets	50 min
C	Design review with the instructor	40 min
D	Build: the first test of the riskiest assumption	40 min
Check by the instructor		30 min

Write each result in `report.md` when you get it.

A team is two lab groups

- ▶ Three or four students.
- ▶ Two XIAOML Kits and two Raspberry Pi 5 boards with the cameras, the cards, and the 27 W supplies.
- ▶ Two or more laptops in the lab network.
- ▶ Parts A to C need no board. Part D uses the boards that the first test needs.

File	Content
proposal.md	The template. Fill it, hand it in.
report.md	The report of today
budget.py	Part B: the budgets
review_sheet.md	Part C: the sheet of the instructor
solutions/	An example proposal and report
TEST_NOTES.md	Test steps for the instructor

Lab files	Labs/day14/ and the lab folders of Days 1 to 13
Software	Python 3.8 or later: budget.py uses only the standard library
The brief	Docs/capstone.md: requirements, rubric, review questions, four examples

Part A: a team and a problem (20 min)

1. **Form the team (5 min).** Join with a second lab group. Write the names and the two group names in `proposal.md` and `report.md`.
2. **Select a problem (10 min).** Read Sections 2 and 7 of `Docs/capstone.md`. Take one example, change one, or write your own problem.
3. **Name the riskiest assumption (5 min).** Which part has the largest chance to fail? Part D tests it.

	Example	XIAO	Raspberry Pi
A	Machine monitor	Vibration model, alarm	Persons, dashboard
B	Voice-controlled camera	Keyword model	Detector for 30 s
C	Room counter with privacy	Person or no person	Count of persons
D	Offline lab assistant	Gesture model	Language model, notes

A solution (weak start)	A problem (good start)
"Run YOLO on a camera"	Nobody should stand near a machine that has an alarm
"Use a language model"	Students need lab answers when the internet is down
"Classify sounds"	A camera should start only when a person asks for it

- ▶ The problem names a user, a place, and a benefit. Then the requirements follow from it.
- ▶ Keep the scope small: reuse the labs of Days 1 to 13 and add one new part.
- ▶ A riskiest assumption is concrete: "the IMU separates a normal fan from a fan with a weight on one blade".

Part B: write the proposal (50 min)

Steps

1. Problem and requirements (15 min)
2. Boards, model, and data (15 min)
3. Budgets with `budget.py` (15 min)
4. The five requirements (5 min)

One page. Write numbers where the template asks for numbers. Mark each value that you did not measure as “estimate”.

Sections of `proposal.md`

- ▶ Problem and user
- ▶ Requirements with numbers
- ▶ Boards and split of the work
- ▶ Model and optimization
- ▶ Data
- ▶ Local decision and telemetry
- ▶ Budgets
- ▶ The five requirements
- ▶ Riskiest assumption and first test

No.	Requirement	Number	Test
R1	Classify the vibration	Recall of “abnormal” at least 0.9	A session that the training does not see
R2	Alarm at the machine	Within 3 s of the start	10 timed events
R3	Alarm with no network	Works with the router off	Switch off the Wi-Fi in the demonstration

- ▶ Rows of the example proposal (machine monitor, solutions/proposal_example.md).
- ▶ Each requirement has a number and a test that you can run before the demonstration on Day 15.
- ▶ “Fast”, “accurate”, and “small” are not requirements. Nobody can check them.

```
python3 budget.py fit --flash-kb 3264 --ram-kb 298.68 --program-kb 300 \  
--params 1534 --bits 8 --peak-values 1028  
python3 budget.py latency window=2000 model=100 decision=400 mqt=100 \  
--limit-ms 3000  
python3 budget.py power --work-mw 217 --rest-mw 19 --work-ms 50 \  
--period-ms 2000 --radio-mw 290.4 --radio-s-per-hour 3 --days 30
```

Command You see (the values of the lecture)

fit	Flash 308 734 of 3 342 336 bytes, RAM 1028 of 305 848 bytes: fits
latency	Total 2600 ms, margin 400 ms (met)
power	24.192 mW, 6.4 days; 30 days need 17.42 Wh (4708 mAh)

Two more commands: data (volume of a message stream) and cost (one site and all sites). Use your own values, and name the source of each input.

Source: Example values of Day 14, Part 3. The program size of 300 KB is an estimate.

Part B: check the five requirements

No.	Requirement	In the machine monitor
1	XIAOML Kit and Raspberry Pi	The XIAO detects the vibration; the Pi sees the persons and runs the dashboard
2	One optimized model, size and accuracy before and after	The XIAO model in <code>float32</code> and <code>int8</code> , recall on the test session
3	One local decision, no internet	The alarm on the XIAO; test with the router off
4	Telemetry with MQTT to a dashboard	The broker on the Pi, the dashboard of Day 13
5	Benchmark table: latency, memory, energy	Both models, with the method of Day 9

If one requirement is not met, change the design now, not on Day 15.

About 8 minutes for each team

1. **Present (3 min):** the problem, the split of the work, the budgets, the riskiest assumption.
2. **Answer (5 min):** the questions of the instructor, with numbers.

While you wait: prepare the answers, then prepare the first test of Part D.

The seven questions

1. A number and a test for each requirement?
2. A value and a source for each budget?
3. Does the model fit: flash, RAM, time?
4. The data: source, size, labels, test set?
5. No network, a failed sensor, a bad update?
6. Which data leaves? Who can send a command?
7. The riskiest assumption and its first test?

- ▶ The instructor fills `review_sheet.md`: the five requirements, the seven questions, the scope, and the result.
- ▶ **You record** in `report.md`: the result, the changes, and the weak points.
- ▶ Make the changes in `proposal.md` before Part D.

Example result (machine monitor)

Approved with changes. (1) The test session of R1 is on a different day or position. (2) The benchmark table needs the measured latency of the XIAO model, not the budget value of 100 ms. Weak point: the battery gives 6.4 days, not 30; the team uses a USB cable.

Key point: A weak point is a result of the review, not a failure of the team.

1. **Plan (5 min):** which board, which lab folder, which data, which number decides.
2. **Run (30 min):** reuse the lab folder of the day that the test needs.
3. **Decide and plan (5 min):** the result, the decision, the plan for Day 15.

Riskiest assumption	First test
A The IMU separates the two states	Two sessions with the Day 2 logger, the Day 3 notebook
B Few false starts in a noisy room	5 minutes of normal speech with the Day 5 model
C The XIAO model works in the room light	20 photos at two times of the day
D Short and fast answers	10 questions, time and length (Day 10)

You record: the test, the result with numbers, the decision (continue or change), and the plan.

Example plan for Day 15 (machine monitor)	Time
1 Record the test session, train the final model	30 min
2 Convert to <code>int8</code> , deploy	30 min
3 Connect the programs of Days 12 and 13	40 min
4 Measure the benchmark table	40 min
5 Rehearse the demonstration with the router off	20 min

- ▶ Day 15 has 100 minutes in the lecture time and 80 minutes in lab Part A for the build. Plan less than 180 minutes.
- ▶ Plan B for the demonstration: a recorded input or a video.

The instructor checks this at the end of the lab:

- ▶ ✓ The proposal meets the five capstone requirements of Docs/capstone.md, Section 2.
- ▶ ✓ Each requirement of the proposal has a number and a test.
- ▶ ✓ Each budget has a value and a source, and each guess is marked as an estimate.
- ▶ ✓ The review sheet says “approved” or “approved with changes”, and the proposal has the changes.
- ▶ ✓ report.md has the result of the first test and a plan for Day 15.

Hand in: proposal.md and report.md.

How do you split the work between the XIAO and the Raspberry Pi in your system? Which budget numbers decided the split, and what did you give up?

A good Decision Log (about 100 words):

- ▶ gives the budget numbers that decided: for example the latency of the main path against its requirement, or the RAM of a model against the RAM of the board,
- ▶ says which numbers are estimates and when you measure them,
- ▶ names what you give up: for example the battery for a cable, or a larger model for a local decision.

Problem	Fix
The idea needs more than one new part	Keep one new part. Reuse the labs for the rest.
No data source	A task with data that you can record today, or a dataset of a lab
Requirement 3 is not met	Move the decision to the XIAO or the Raspberry Pi
A budget has no source	Write “estimate”, and measure it in Part D or on Day 15
<code>budget.py</code> says <code>is not name=value</code>	Write <code>window=2000</code> , with no space
The first test needs more than 30 minutes	Test only the assumption, with the smallest data that shows it

The README of the lab has the steps and the expected values.

This lecture adapts material from these sources:

- ▶ *Machine Learning Systems* by Vijay Janapa Reddi and contributors, Harvard University. mlsysbook.ai, CC BY-NC-SA 4.0.
- ▶ *XIAO: Big Power, Small Board* by Lei Feng and Marcelo Rovai. github.com/Mjrovai/XIAO_Big_Power_Small_Board-ebook, GPL-3.0.

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